TRACE: Trajectory Recovery for Continuous Mechanism Evolution in Causal Representation Learning

CS 412 Intermediate Project Report

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ABSTRACT

Temporal causal representation learning (CRL) methods assume that causal mechanisms switch instantaneously between a few discrete domains, yet real systems often change *continuously*. A vehicle's dynamics evolve gradually through a turn, and human gait shifts smoothly from walking to running. We model each intermediate mechanism as a convex combination of finitely many *atomic mechanisms* with time-varying mixing coefficients $\boldsymbol{\alpha}(t)$, and ask whether both the latent causal variables and the trajectory $\boldsymbol{\alpha}(t)$ can be recovered from observations. So far we have formalized the problem, proved that the latent variables are identifiable and that $\boldsymbol{\alpha}(t)$ admits a pointwise recovery bound, and built TRACE, a two-stage Mixture-of-Experts framework that learns one atomic mechanism per expert and recovers $\boldsymbol{\alpha}(t)$ at test time by least-squares projection. On synthetic data TRACE reaches a trajectory correlation of 0.94 versus 0.72 for the strongest discrete-switching baseline, the theoretical bound holds empirically, and on pixel-based CartPole it reaches MCC 0.970 and trajectory correlation 0.95. The remaining work covers a trajectory-level guarantee that exploits temporal smoothness, real-world evaluation on vehicle and human-motion data, and an ablation of the model's capacity limit.

1 INTRODUCTION

Causal representation learning (CRL) [15, 21] seeks to uncover latent causal variables and their relationships from high-dimensional observations. Without additional structure this problem is not identifiable [13], so temporal CRL methods exploit time-delayed dependencies and distribution changes across domains [6, 7, 9]. TDRL [19] achieves identifiability through time-delayed causal influences, and NCTRL [18] extends it to nonstationary data with unknown discrete switches. Later work addresses instantaneous effects [11, 12], non-invertible mixing [2] and sparse transitions [17]. All of these methods share one assumption, namely that the mechanism takes values in a finite set and switches instantaneously.

In real systems mechanisms typically evolve continuously. A turning maneuver gradually shifts the causal relations between steering, velocity and lateral acceleration, and the walk-to-run transition [3, 5] continuously reorganizes limb coordination. Tracking such transitions matters in practice because the critical points along the way are where prediction, anomaly detection or intervention is most valuable.

Our key observation is that the space of mechanisms often has low-dimensional structure. Just as any color is a mixture of RGB primaries, or a document a mixture of topics [1], an intermediate causal mechanism can be written as a convex combination of a few *atomic mechanisms*. Each mechanism state is then a point in a simplex whose vertices are learned from pure-domain data (Figure 1). This raises the central question of the project: **how can we learn representations that capture the continuous evolution of causal mechanisms?** Two challenges follow. We need a latent space in which mechanism-specific structure is identifiable, and we need to recover the mixing trajectory $\boldsymbol{\alpha}(t)$ from unlabeled test sequences.

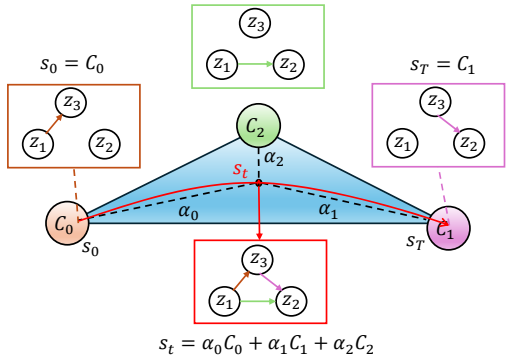


Figure 1: Continuous mechanism transitions as trajectories through a simplex. Atomic mechanisms $\{C_0, C_1, C_2\}$ are vertices, and an interior point s_t is a convex combination inducing a mixed causal graph. The red and dashed curves are distinct trajectories between the same endpoints, which discrete-switching models cannot distinguish.

Contributions so far. (1) We formalize CRL with continuously evolving mechanisms as convex combinations of atomic mechanisms. (2) We prove that the latent variables are identifiable up to permutation and component-wise transformation, give a verifiable sufficient condition, and prove a pointwise error bound for recovering $\boldsymbol{\alpha}(t)$. (3) We propose TRACE, a two-stage framework that uses a Mixture-of-Experts [8, 16] to learn atomic mechanisms and least-squares projection to recover $\boldsymbol{\alpha}(t)$. (4) We validate TRACE on synthetic data and on pixel-based CartPole.

2 PROBLEM DESCRIPTION

Generative model. Let $\mathbf{z}_t \in \mathbb{R}^d$ be latent causal variables and $\mathbf{x}_t \in \mathbb{R}^p$ the observation,

$$\mathbf{z}_t = \sigma\big(W(t)\,\mathbf{z}_{t-1}\big) + \boldsymbol{\epsilon}_t, \qquad \mathbf{x}_t = g(\mathbf{z}_t), \tag{1}$$

where g is an invertible nonlinear mixing function, σ a component-wise activation (e.g. LeakyReLU) and $\boldsymbol{\epsilon}_t$ independent noise. Prior

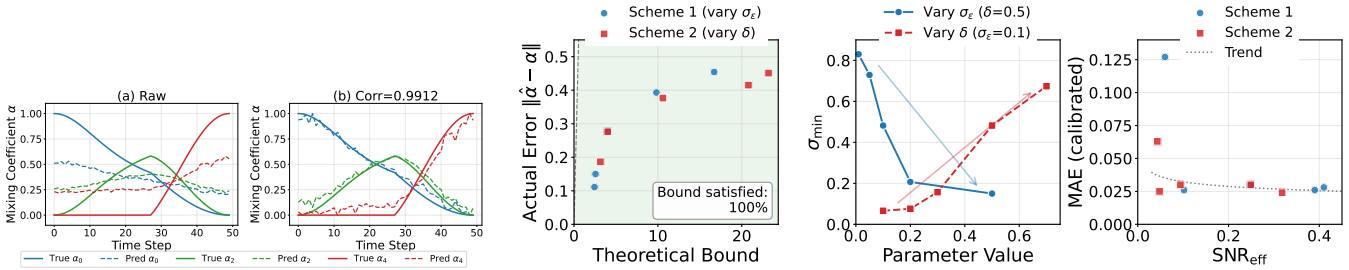


Figure 2: Left: recovery of $\boldsymbol{\alpha}(t)$ with $K_{\text{active}} = 3$ (domains 0, 2, 4), Corr. 0.99 after calibration. Right: empirical check of Theorem 2, with recovery error against the theoretical bound (all points below $y = x$), $\sigma_{\min}$ under varying noise and perturbation, and MAE against the effective SNR.

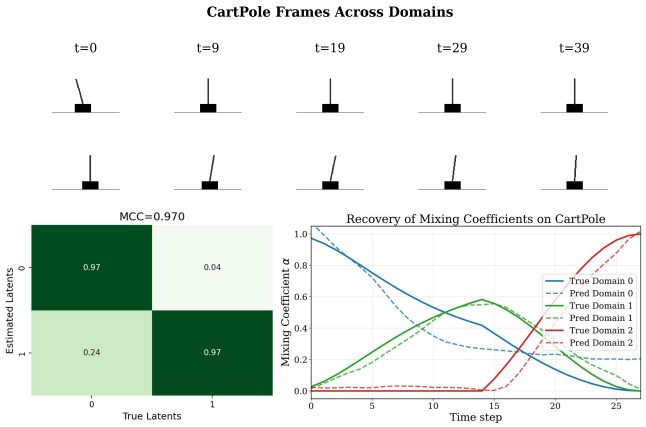


Figure 3: Modified CartPole. Top: frames from different domains. Bottom left: correlation matrix of recovered latents (MCC 0.970). Bottom right: recovered mixing coefficients (Corr. 0.95).

work assumes $W(t) \in \{W^{(0)}, \ldots, W^{(K-1)}\}$. We instead let

$$W(t) = \sum_{k=0}^{K-1} \alpha_k(t)\, W^{(k)}, \qquad \boldsymbol{\alpha}(t) \in \Delta^{K-1}, \tag{2}$$

so the causal graph moves continuously inside the simplex spanned by K atomic mechanisms.

Assumption 1 (Distinguishable mechanisms)**.** No $W^{(k)}$ is a convex combination of the others. Concretely, for each latent dimension i the rows $\{(W^{(k)} - W^{(0)})_{i,:}\}_{k=1}^{K-1}$ are linearly independent, which implies the capacity limit $K \le d + 1$.

Task. During training we observe labeled trajectories from the K pure domains ($\boldsymbol{\alpha}$ one-hot). At test time we observe an unlabeled trajectory $\mathbf{x}_{1:T}$ whose $\boldsymbol{\alpha}(t)$ varies smoothly, with neither latents nor labels available. The goal is (i) latents $\hat{\mathbf{z}}_t$ identifiable up to permutation and component-wise transformation, and (ii) the trajectory $\{\boldsymbol{\alpha}(t)\}_{t=1}^{T}$.

Identifiability results.

Theorem 1 (Latent variables)**.** *If g is invertible and $p(z_{i,t} \mid \mathbf{z}_{t-1})$ varies sufficiently across the K domains, any $\hat{\mathbf{z}}_t$ satisfying the conditional-independence constraints obeys $\hat{z}_i = h_i(z_{\pi(i)})$ for a permutation π and strictly monotone h_i.*

This builds on TDRL [19], which needs an abstract "sufficient variability" condition. We show that Assumption 1 implies it. Under Gaussian noise,

$$\frac{\partial^2 \log p(z_{i,t} \mid \mathbf{z}_{t-1}, k)}{\partial z_{i,t}\, \partial \mathbf{z}_{t-1}} = \frac{\sigma'([W^{(k)}\mathbf{z}_{t-1}]_i)}{\sigma_i^2}\, (W^{(k)})_{i,:}^{\top},$$

so cross-domain differences are, to first order, proportional to the rows of $W^{(k)} - W^{(0)}$ and inherit their linear independence when $\sigma' \neq 0$.

Let $\hat{\boldsymbol{\mu}}^{(k)}$ be the conditional mean of $\hat{\mathbf{z}}_t$ under domain k, and $\hat{B} = [\hat{\boldsymbol{\mu}}^{(1)} - \hat{\boldsymbol{\mu}}^{(0)}, \ldots, \hat{\boldsymbol{\mu}}^{(K-1)} - \hat{\boldsymbol{\mu}}^{(0)}]$ with smallest singular value $\sigma_{\min} > 0$.

Theorem 2 (Pointwise recovery)**.** *Under Theorem 1, $\hat{\boldsymbol{\alpha}}(t) = \hat{B}^{\dagger}(\hat{\mathbf{z}}_t - \hat{\boldsymbol{\mu}}^{(0)})$ satisfies*

$$\|\hat{\boldsymbol{\alpha}}(t) - \boldsymbol{\alpha}(t)\| \le \frac{1}{\sigma_{\min}}\big(\|\hat{\boldsymbol{\epsilon}}_t\| + \delta_{\text{approx}}\big), \qquad \delta_{\text{approx}} = O(\epsilon^2),$$

where $\epsilon = \max_k \|W^{(k)} - W^{(0)}\|$.

Proof sketch. By Theorem 1, $\hat{\mathbf{z}} = P\,h(\mathbf{z})$. Expanding h around $\boldsymbol{\mu}^{(0)}$ gives $\hat{B} = PDB$ with $D = \mathrm{diag}(h_i'(\mu_i^{(0)}))$ invertible, so $\hat{B}$ has full column rank and $\hat{B}^{\dagger}\hat{B} = I$. Writing $\hat{\mathbf{z}}_t - \hat{\boldsymbol{\mu}}^{(0)} = \hat{B}\boldsymbol{\alpha} + r(\boldsymbol{\alpha}) + \hat{\boldsymbol{\epsilon}}_t$ with residual $\|r\| \le \delta_{\text{approx}}$, we get $\hat{\boldsymbol{\alpha}} - \boldsymbol{\alpha} = \hat{B}^{\dagger}(r + \hat{\boldsymbol{\epsilon}}_t)$, and $\|\hat{B}^{\dagger}\| = 1/\sigma_{\min}$. A second-order Taylor expansion of f in W and of h gives $\delta_{\text{approx}} = O(\epsilon^2)$. □

The bound also predicts a *geometric bottleneck*. As K approaches $d + 1$ the basis columns become nearly collinear, $\sigma_{\min}$ shrinks, and recovering $\boldsymbol{\alpha}$ degrades even when every atomic mechanism is learned correctly.

Method (TRACE).. TRACE has two stages (Figure 4). **Stage 1** extends a sequential VAE [10] with an MoE prior. A shared encoder maps $(\mathbf{x}_t, \mathbf{x}_{t-L:t-1})$ to a Gaussian posterior, reflecting that the causal variables are shared while their relations change. Expert k is a normalizing flow [14] that maps $\mathbf{z}_t$ to independent noise given its history, giving the transition prior $p_k(\mathbf{z}_t \mid \mathbf{z}_{t-L:t-1})$. One-hot gating routes each training sample to its domain's expert, and we maximize $\mathbb{E}_q[\log p(\mathbf{x}_t \mid \mathbf{z}_t) + \log p_k(\mathbf{z}_t \mid \mathbf{z}_{t-L:t-1}) - \log q(\mathbf{z}_t \mid$

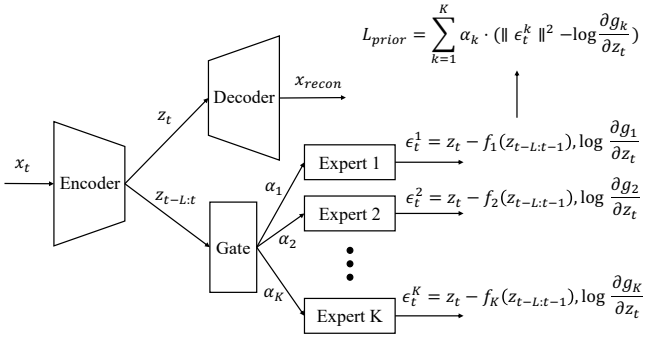


Figure 4: TRACE. Stage 1 trains a shared encoder with one-hot gating to domain-specific experts. Stage 2 recovers $\boldsymbol{\alpha}$ by least-squares projection.

$\mathbf{x}_t)$]. **Stage 2** computes $\hat{\boldsymbol{\mu}}^{(k)}$ on validation data of each domain, forms $\hat{B}$, encodes a test observation by its posterior mean, and applies $\hat{\boldsymbol{\alpha}}(t) = \mathrm{Proj}_{\Delta^{K-1}}(\hat{B}^{\dagger}(\mathbf{z}_t - \hat{\boldsymbol{\mu}}^{(0)}))$ followed by a moving average with window $2w+1$. No transition data are needed for training, so the method generalizes to intermediate states never observed.

3 DATA

Synthetic dynamical system. $d = 8$ latents with second-order dynamics

$$\mathbf{z}_t = \sigma\big(\sigma(\mathbf{z}_{t-1}W^{(k)}) + \sigma(\mathbf{z}_{t-2}W_{\mathrm{lag2}})\big) + \boldsymbol{\epsilon}_t,$$

LeakyReLU slope 0.2 and $\boldsymbol{\epsilon}_t \sim \mathcal{N}(0, 0.1^2 I)$. We build $K = 5$ atomic mechanisms $W^{(k)} = W_{\mathrm{base}} + \delta W^{(k)}$, where each $\delta W^{(k)}$ modifies one distinct off-diagonal edge, which satisfies Assumption 1. Observations $\mathbf{x}_t \in \mathbb{R}^{16}$ come from an invertible orthogonal MLP. Training uses 40,000 pure-domain trajectories per domain (200,000 total) of length 50. Test trajectories mix $K_{\mathrm{active}} = 3$ domains (e.g. $0 \to 2 \to 4$) with linear or sinusoidal schedules, so ground-truth $\boldsymbol{\alpha}(t)$ is known.

Modified CartPole. A semi-synthetic system with latent state (x, v, θ, ω) rendered as 128×128 grayscale frames. Five domains each strengthen one causal edge (e.g. $v \to \theta$, $\theta \to \omega$), with constant noise variance across domains. This is a setting where NCTRL's identifiability conditions fail but ours hold. We use 1,000 trajectories of length 100 per domain.

Real-world data. UAVDT [4] provides UAV traffic video with annotated vehicle boxes. We extract turning trajectories at intersections (length $\geq$ 20 frames, direction change $\geq 60^\circ$) and train on straight horizontal and vertical motion as the two atomic mechanisms. *CMU Motion Capture* provides 31-joint skeletons at 120 Hz. We use walk-to-run trials from six subjects (over 1,000 transition frames, $p = 188$ features from joint positions, velocities and hip statistics) and train on pure walking and running. Neither dataset has ground-truth $\boldsymbol{\alpha}(t)$, so we evaluate against physically motivated proxies (velocity direction for vehicles, normalized hip speed for gait).

Table 1: Synthetic data. MCC is computed on pure-domain test data, Corr. on continuous transitions. NA means the method needs discrete labels and cannot output weights during transitions.

Method	Weight Corr. ↑	Latent MCC ↑
NCTRL (hard)	0.67 ± 0.03	0.90 ± 0.09
NCTRL (soft)	0.72 ± 0.01	0.90 ± 0.09
TDRL	NA	0.89 ± 0.03
LEAP	NA	0.66 ± 0.05
iVAE	NA	0.53 ± 0.02
PCL	NA	0.59 ± 0.04
TRACE	**0.94 ± 0.05**	**0.96 ± 0.03**

4 WHAT WE HAVE DONE SO FAR

Completed work.

(1) **Problem formulation.** Continuous mechanisms as convex combinations of atomic mechanisms (Eqs. 1–2), with the distinguishability assumption and its capacity limit $K \leq d + 1$.
(2) **Theory.** Proofs of latent identifiability (Theorem 1), of the bridge from Assumption 1 to sufficient variability, of pointwise recovery of $\boldsymbol{\alpha}(t)$ (Theorem 2) and of $\delta_{\mathrm{approx}} = O(\epsilon^2)$.
(3) **Implementation.** The full TRACE pipeline (Stage 1 MoE sequential VAE and Stage 2 least-squares solver), the data generators for the synthetic system and modified CartPole, and the six baselines.
(4) **Experiments.** Synthetic comparison with baselines, empirical validation of Theorem 2, and CartPole from pixels.

Implementation details. The model is implemented in PyTorch. For synthetic data the encoder and decoder are 3-layer MLPs (hidden size 128), $d = 8$, lag $L = 2$, and each expert is a normalizing-flow transition prior. We train with Adam (learning rate 5×10^{-4}, batch size 64, 100 epochs) with KL weight $\beta = 2 \times 10^{-3}$ and prior weight $\gamma = 2 \times 10^{-2}$. For CartPole the encoder is a CNN (three stride-2 convolutions with 32, 64 and 128 channels followed by FC 256 and FC 4), trained for 200 epochs. Synthetic training takes 6–8 hours on one A100 and CartPole 15–20 hours on two A100s. At test time we use window $w = 5$ and two-point scale calibration from the trajectory endpoints.

Metrics and baselines. Latent identifiability is measured by the mean correlation coefficient (MCC) [6], and trajectory recovery by the Pearson correlation between true and recovered $\boldsymbol{\alpha}(t)$ (Corr.). We compare with NCTRL [18] (hard and soft gating), TDRL [19], LEAP [20], iVAE [9] and PCL [7]. All baselines assume discrete mechanisms.

Synthetic results. Table 1 shows that TRACE reaches the highest trajectory correlation (0.94 versus 0.72 for NCTRL-soft) and the highest latent MCC (0.96). Figure 2 (left) shows a three-domain transition. After two-point scale calibration the recovered trajectory tracks both the sequential activation and the timing of the intermediate peak (Corr. 0.99).

Validating the theory. We vary the noise level and the perturbation size (Table 2). Three observations match the theory. Corr.

Table 2: Checking the predictions of Theorem 2. MAE before (raw) and after (cal) scale calibration; $\mathrm{SNR}_{\mathbf{eff}} = \sigma_{\min}/(\bar{\epsilon} + \delta_{\mathrm{approx}})$.

Vary noise σ_ϵ ($\|W^{(k)} - W^{(0)}\|_F = 0.5$)					
σ_ϵ	0.01	0.05	0.10	0.20	0.50
MCC	0.996	0.994	0.984	0.849	0.703
Corr.	0.998	0.998	0.998	0.996	0.854
$\mathrm{MAE}_{\mathrm{raw}}$	0.077	0.106	0.254	0.278	0.321
$\mathrm{MAE}_{\mathrm{cal}}$	0.028	0.026	0.030	0.026	0.127
Vary perturbation ($\sigma_\epsilon = 0.1$)					
$\|W^{(k)} - W^{(0)}\|_F$	0.1	0.2	0.3	0.5	0.7
MCC	0.934	0.953	0.970	0.984	0.990
Corr.	0.972	0.996	0.997	0.998	0.998
$\mathrm{MAE}_{\mathrm{raw}}$	0.319	0.292	0.266	0.254	0.194
$\mathrm{MAE}_{\mathrm{cal}}$	0.063	0.025	0.030	0.030	0.024

stays above 0.97 even when MCC degrades, so the trajectory shape is preserved. Calibration reduces MAE by 3–10×, consistent with the bias term $\hat{B}^{\dagger} r(\boldsymbol{\alpha})$ acting as a near-linear distortion. Both MCC and MAE improve as the perturbation grows, as predicted by the $1/\sigma_{\min}$ dependence. Figure 2 (right) confirms that every recovery error lies below the bound of Theorem 2.

CartPole from pixels. Figure 3 shows that TRACE recovers the observable latents (x, θ) from 128×128 frames with MCC 0.970 and recovers the three-domain trajectory with Corr. 0.95, so the framework extends from vectors to images.

5 WHAT REMAINS TO BE DONE

(1) **Trajectory-level theory (by Nov 20).** Theorem 2 treats each time step independently, so its bound does not improve with T. We will exploit the smoothness of $\boldsymbol{\alpha}(t)$ (bounded total variation) through a regularized estimator and derive a finite-sample error rate in T, with complete proofs.
(2) **Real-world experiments (by Nov 15).** Finish the UAVDT vehicle-turning and CMU walk-to-run experiments, comparing TRACE with NCTRL against the proxy trajectories.
(3) **Capacity ablation (by Nov 22).** Train with $K_{\mathrm{total}} = 10$ and evaluate $K_{\mathrm{active}} \in \{2, \ldots, 7\}$ on sequential, overlapping and oscillating schedules, reporting recovery of both $\boldsymbol{\alpha}(t)$ and $W(t)$, to test the geometric bottleneck predicted by Theorem 2.
(4) **Robustness (by Nov 25).** Smoothing window size, a Kolmogorov–Smirnov test of the convex-interpolation assumption, and out-of-distribution transitions.
(5) **Deliverables.** Final presentation (mid November) and the 8–10 page final report with full proofs (December 1).

TEAM CONTRIBUTIONS

Shicheng Fan and Jie Yang ran the experiments. All three members discussed the theoretical framework and the proofs together. Honghui Bao wrote this report.

CODE AVAILABILITY

Code, data generators and reports are maintained at https://github.com/HonghuiBao2000/cs412-trace.

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